

Notations: $\mathcal{F}(X)$ will denote the set of all fuzzy sets on an $X \neq \emptyset$. $A, B \in \mathcal{F}(X)$.

1. **Fuzzy Partition:** Consider the terms *High*, *Medium*, *Low*, *Very Low* when being referred to **Pressure** in a gas chamber. By suitably assuming a domain for **Pressure**,
 - Construct a Ruspini partition of the domain with fuzzy sets that according to you capture the corresponding concepts.
 - Can you do the above with fuzzy sets that are not convex?
 - Discuss the ordering between the fuzzy sets you have constructed above.

2. **Cardinality of a fuzzy set:** The sigma count of A is given as $|A| = \sum_{x \in X} A(x)$.

- Does the above formula capture the notion of cardinality of a set?
- What properties of cardinality of a classical set does the above preserve?
- For instance, is the following equality true: $|A| + |B| = |A \cup B| + |A \cap B|$?

3. **Measure of Subsethood:** We have seen at least two different ways of defining subsethood / ordering among fuzzy sets. The following generalises even this concept to a degree:

Let X be finite and $A, B \in \mathcal{F}(X)$. The *degree of subsethood* of A in B is defined as

$$S(A, B) = \frac{1}{|A|} \left(|A| - \sum_{x \in X} \max[0, A(x) - B(x)] \right).$$

(a) Is it true that $0 \leq S(A, B) \leq 1$?

(b) Show that $S(A, B) = \frac{|A \cap B|}{|A|}$.

4. Let $[A]_\alpha$ denote the α -cut of A for an $\alpha \in [0, 1]$, i.e., $[A]_\alpha = \{x \in X \mid A(x) \geq \alpha\}$. Which of the following are true for every $\alpha \in [0, 1]$:

(a) $[A \cap B]_\alpha = A_\alpha \cap B_\alpha$

(b) $[A \cup B]_\alpha = A_\alpha \cup B_\alpha$

(c) $A_\alpha = \bigcup_{\beta < \alpha} A_\beta$.

5. For an α -cut $[A]_\alpha$ of A let

$$\chi_{[A]_\alpha} = \begin{cases} 1, & x \in [A]_\alpha, \\ 0, & \text{otherwise,} \end{cases}$$

be its characteristic function. Show that

$$A = \bigcup_{\alpha \in \Lambda} \alpha \cdot \chi_{[A]_\alpha},$$

i.e., A can be decomposed in terms of its α -cuts.

6. **Convexity of a Fuzzy Set:** We know that a fuzzy set $A \in \mathcal{F}(\mathbb{R})$ is convex if for every $\alpha \in [0, 1]$, $[A]_\alpha \subset \mathbb{R}$ is convex. Show that the following is an equivalent way of characterising the same concept:

$$A \text{ is convex iff } A(\lambda x_1 + (1 - \lambda)x_2) \geq \min[A(x_1), A(x_2)],$$

where $x_1, x_2 \in \mathbb{R}$ and $\lambda \in [0, 1]$.

7. Let $N : [0, 1] \rightarrow [0, 1]$ be an involutive fuzzy negation, i.e., $N(N(x)) = x$ for all $x \in [0, 1]$. Is it true that such a negation N is also continuous?